

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

Digital Forensics Lab 3 - Data Carving with XXD, Binwalk and Scalpel

Lab Title:	Data Carving with XXD, Binwalk and Scalpel
Student Name:	Yusuf Oluwaseun Dada
Student ID:	2025/FWSD/11339
Course Code:	SBT-DF202
Course Name:	Computer and Digital Forensics

Instructor: Mr. Aminu Idris, AMCPN

Date: 05/09/2026

Lab Setup: Tools and Workspace

```
(kali@kali)-[~/Desktop]
$ sudo apt update
sudo apt install -y steghide stegseek
which steghide stegseek sha256sum xxd diff strings file
[sudo] password for kali:
Get:1 https://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 https://kali.download/kali kali-rolling/main amd64 Packages [21.4 MB]
Get:3 https://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.3 MB]
Get:4 https://kali.download/kali kali-rolling/contrib amd64 Packages [114 kB]
Get:5 https://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [264 kB]
Get:6 https://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:7 https://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Fetched 76.2 MB in 1min 29s (856 kB/s)
85 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  steghide  stegseek

Installing dependencies:
  libmbedtls

Suggested packages:
  libmbedtls-dev  mpcrypt

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 85
  Download size: 355 kB
  Space needed: 1,133 kB / 50.5 GB available

Get:1 https://kali.download/kali kali-rolling/main amd64 libmbedtls amd64 2.5.8-8+b2 [71.8 kB]
Get:2 https://kali.download/kali kali-rolling/main amd64 steghide amd64 0.5.1+git20240220-2 [154 kB]
Get:3 https://kali.download/kali kali-rolling/main amd64 stegseek amd64 0.6+git20210910.ff677b9-2 [130 kB]
Fetched 355 kB in 2s (173 kB/s)
```

```
Session  Actions  Edit  View  Help

Unpacking libmbedtls:amd64 (2.5.8-8+b2)...
Selecting previously unselected package steghide.
Preparing to unpack ./steghide_0.5.1+git20240220-2_amd64.deb...
Unpacking steghide (0.5.1+git20240220-2)...
Selecting previously unselected package stegseek.
Preparing to unpack ./stegseek_0.6+git20210910.ff677b9-2_amd64.deb...
Unpacking stegseek (0.6+git20210910.ff677b9-2)...
Setting up libmbedtls:amd64 (2.5.8-8+b2)...
Setting up stegseek (0.6+git20210910.ff677b9-2)...
Setting up steghide (0.5.1+git20240220-2)...
Processing triggers for libc-bin (2.42-17)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.3.2)...
Scanning processes ...
Scanning linux images ...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
/usr/bin/steghide
/usr/bin/stegseek
/usr/bin/sha256sum
/usr/bin/xxd
diff: aliased to diff --color=auto
/usr/bin/strings
/usr/bin/file
```

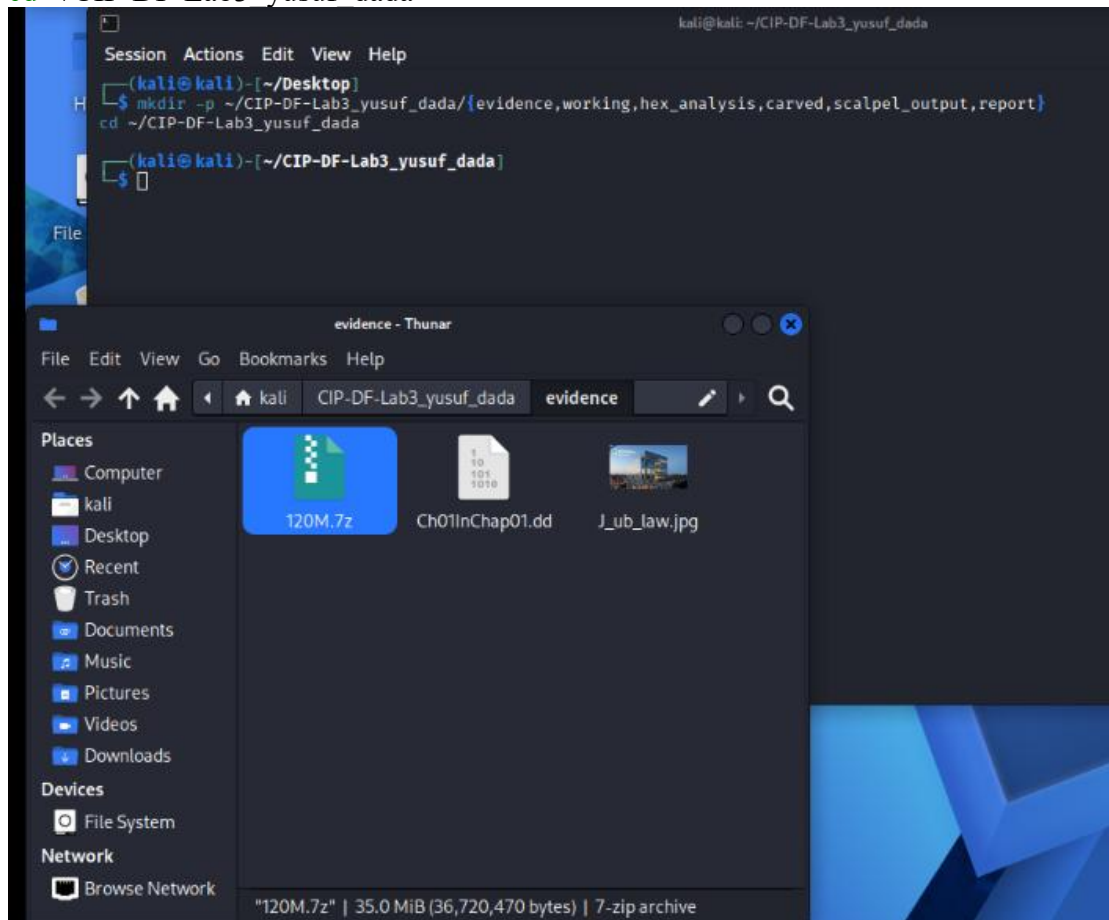
sudo apt update

sudo apt install -y xxd binwalk scalpel p7zip-full sleuthkit

which xxd strings binwalk scalpel 7z img_stat mmls fsstat fls icat blkcat md5sum sha256sum

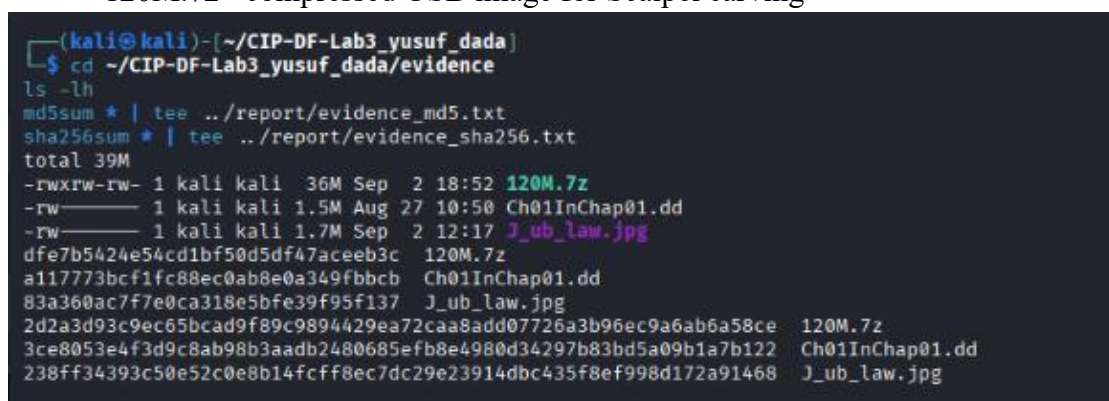
Lab Folder Creation

```
mkdir -p ~/CIP-DF-Lab3_yusuf_dada/{evidence,working,hex_analysis,carved,scalpel_output,report}
cd ~/CIP-DF-Lab3_yusuf_dada
```



The three required files from my course portal has been moved into evidence/:

- Ch01InChap01.dd - the same forensic DD image from your earlier Sleuth Kit lab
- J_ub_law.jpg - JPEG sample for hex/signature analysis
- 120M.7z - compressed USB image for Scalpel carving



```
cd ~/CIP-DF-Lab3_Firstname_Lastname/evidence
```

```
ls -lh
```

```
md5sum * | tee ../report/evidence_md5.txt
```

```
sha256sum * | tee ../report/evidence_sha256.txt
```

Part A - File Signatures and Hex Analysis

A1. Identification of the JPEG header (SOI marker)

```
(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada/evidence]
$ cd ~/CIP-DF-Lab3_yusuf_dada
xxd evidence/J_ub_law.jpg | head -n 5
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00 .....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013 .....@.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201 .....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100 .....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400 .....
```

```
cd ~/CIP-DF-Lab3_yusuf_dada
xxd evidence/J_ub_law.jpg | head -n 5
```

The first byte shows a valid JPEG that starts with ff d8 ff (the Start of Image / SOI marker)

A2. Identification of the JPEG footer (EOI marker)

```
(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada]
$ xxd evidence/J_ub_law.jpg | tail -n 5
0019d1b0: 294f ae4a a859 c9b4 2885 515a f676 3a80 )0.J.Y..(.QZ.v:.
0019d1c0: fc7f 81df 1c79 f5e1 dc0f 53e3 a4d5 5f45 .....y....S..._E
0019d1d0: 182a d52d 2406 abc1 ad69 9479 0789 d1e1 ..*.-$....i.y....
0019d1e0: ff00 5254 0938 ce40e 6ff5 f75a 9a54 f565 ..RT.8...o..Z.T.e
0019d1f0: d351 9ebf ffd9 .Q....
```

```
xxd evidence/J_ub_law.jpg | tail -n 5
```

A valid JPEG always ends with ff d9 (End of Image / EOI marker). and it has been confirmed

It shows both header and footer hex views, with the SOI (ffd8ff) and EOI (ffd9) bytes visibly highlighted

A3. Creation of a plain hexadecimal dump

```
(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada]
$ xxd evidence/J_ub_law.jpg > hex_analysis/J_ub_law_hexdump.txt
wc -l hex_analysis/J_ub_law_hexdump.txt
head -n 10 hex_analysis/J_ub_law_hexdump.txt

105760 hex_analysis/J_ub_law_hexdump.txt
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00 .....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013 .....@.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201 .....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100 .....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400 .....
00000050: 0000 1001 0200 0900 0000 e600 0000 1201 .....
00000060: 0300 0100 0000 0100 0000 1501 0300 0100 .....
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00 .....
00000080: 0000 1b01 0500 0100 0000 f700 0000 2801 .....(
00000090: 0300 0100 0000 0200 0000 3101 0200 2100 .....1...!
```

```
xxd evidence/J_ub_law.jpg > hex_analysis/J_ub_law_hexdump.txt
wc -l hex_analysis/J_ub_law_hexdump.txt
head -n 10 hex_analysis/J_ub_law_hexdump.txt
```


A4. Reconstruct the image using reverse-hexdump

```
(kali@kali)~[/CIP-DF-Lab3_yusuf_dada]
$ xxd -r hex_analysis/J_ub_law_hexdump.txt > working/J_ub_law_reconstructed.jpg
file working/J_ub_law_reconstructed.jpg

working/J_ub_law_reconstructed.jpg: JPEG image data, Exif standard: [TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=206,
PhotometricInterpretation=RGB, manufacturer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left], baseline, precision 8, 1920x1080, components
3
```

```
xxd -r hex_analysis/J_ub_law_hexdump.txt > working/J_ub_law_reconstructed.jpg
file working/J_ub_law_reconstructed.jpg
```

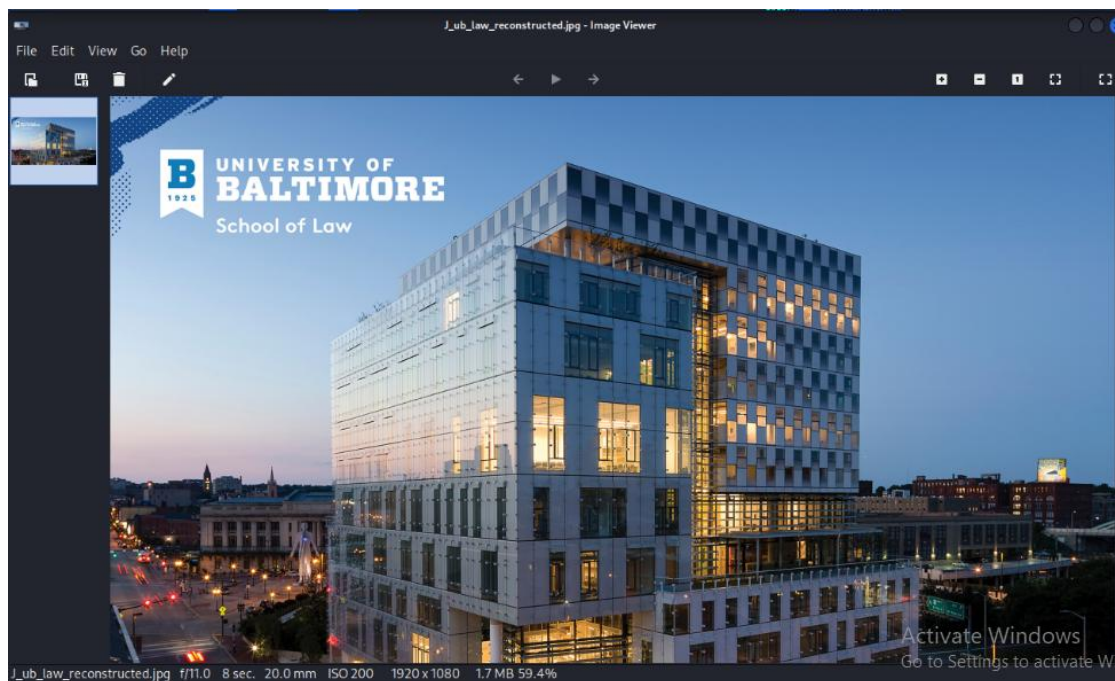
A5. Verify the reconstruction with hash comparison

```
(kali@kali)~[/CIP-DF-Lab3_yusuf_dada]
$ md5sum evidence/J_ub_law.jpg working/J_ub_law_reconstructed.jpg
sha256sum evidence/J_ub_law.jpg working/J_ub_law_reconstructed.jpg
83a360ac7f7e0ca318e5bfe39f95f137 evidence/J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137 working/J_ub_law_reconstructed.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 evidence/J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 working/J_ub_law_reconstructed.jpg
```

```
md5sum evidence/J_ub_law.jpg working/J_ub_law_reconstructed.jpg
sha256sum evidence/J_ub_law.jpg working/J_ub_law_reconstructed.jpg
```

Both hash comparisons showing the original and reconstructed file hashes **matching exactly** - xxd/xxd, so identical hashes here prove the reconstruction preserved every byte perfectly.

Visual confirmation:



```
xdg-open working/J_ub_law_reconstructed.jpg 2>/dev/null || eog working/J_ub_law_reconstructed.jpg
```

Part B - Examine Embedded Metadata-Like Content

```
(kali@kali)~[/CIP-DF-Lab3_yusuf_dada]
$ cd ~/CIP-DF-Lab3_yusuf_dada
strings evidence/J_ub_law.jpg | head -n 40
echo "---- Searching for common metadata indicators ----"
strings evidence/J_ub_law.jpg | grep -iE "canon|nikon|sony|samsung|apple|adobe|photoshop|gimp|software|camera|date|gps|exif" | tee hex_analysis/metadata_strings_found.txt
echo "---- Full readable string count ----"
strings evidence/J_ub_law.jpg | wc -l
Exif
NIKON CORPORATION
NIKON D4
Adobe Photoshop 21.1 (Macintosh)
2020:08:20 10:20:05
HOWARD KORN
0221
2013:10:08 18:56:21
2013:10:08 18:56:21
2027750
17.0-35.0 mm f/2.8
Adobe_CM
Adobe
b34r
7Gwgw
AQaq
dEU6te
7Gwgw
2VtWX
I;[h
[$wik
S_b[
Suhf
KC[ LO
```

```
Y{h}
{"v>
04YC[hakK
X-QI:
I2V:$C
VdMh/
;/nM
5CuW
ofEO'
Nmon=W
jv;g
b"ki
h!6YXk^
us\}
's'gG
---- Searching for common metadata indicators ----
Exif
NIKON CORPORATION
NIKON D4
Adobe Photoshop 21.1 (Macintosh)
Adobe_CM
Adobe
"Photoshop 3.0
PgPenum
PgPs
Adobe_CM
Adobe
http://ns.adobe.com/xap/1.0/
" id="WSMMMPcehiH2reS2NtCzk9d"?> <x:xmpmeta xmlns:x="adobe:meta/" x:xmpptk="Adobe XMP Core 6.0-c002 79.164352, 2020/01/30-15:50:38" > <rdf:
RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"> <rdf:Description rdf:about="" xmlns:aux="http://ns.adobe.com/exif/1.0/aux/" xmlns:dc="ht
tp://purl.org/dc/elements/1.1/" xmlns:photoshop="http://ns.adobe.com/photoshop/1.0/" xmlns:xmp="http://ns.adobe.com/xap/1.0/" xmlns:xmpMM="http://ns.
adobe.com/xap/1.0/mm/" xmlns:stRef="http://ns.adobe.com/xap/1.0/sType/ResourceRef#" xmlns:stEvt="http://ns.adobe.com/xap/1.0/sType/ResourceEvent#" xm
```

```
/jpeg to application/vnd.adobe.photoshop"/> <rdf:li stEvt:action="saved" stEvt:instanceID="xmp.iid:981ec349-e69e-479f-b5d4-600905e0b0e6" stEvt:when="
2020-07-02T10:21:18-04:00" stEvt:softwareAgent="Adobe Photoshop 21.1 (Macintosh)" stEvt:changed="/" /> <rdf:li stEvt:action="saved" stEvt:instanceID="
xmp.iid:0bb33b69-4279-4f45-afce-49859bf97ea8" stEvt:when="2020-08-17T14:09:42-04:00" stEvt:softwareAgent="Adobe Photoshop 21.1 (Macintosh)" stEvt:cha
nged="/" /> <rdf:li stEvt:action="converted" stEvt:parameters="from application/vnd.adobe.photoshop to image/jpeg"/> <rdf:li stEvt:action="derived" st
Evt:parameters="converted from application/vnd.adobe.photoshop to image/jpeg"/> <rdf:li stEvt:action="saved" stEvt:instanceID="xmp.iid:3f0c89f0-f710-
45fe-a8f8-01d5e14cc22f" stEvt:when="2020-08-17T14:09:42-04:00" stEvt:softwareAgent="Adobe Photoshop 21.1 (Macintosh)" stEvt:changed="/" /> <rdf:li stE
vt:action="saved" stEvt:instanceID="xmp.iid:9e7f6b27-04e9-43e7-ae79-c3f2fb9c2ac9" stEvt:when="2020-08-20T10:20:05-04:00" stEvt:softwareAgent="Adobe P
hotoshop 21.1 (Macintosh)" stEvt:changed="/" /> </rdf:Seq> </xmpMM:History> <crs:ToneCurvePV2012> <rdf:Seq> <rdf:li>0, 0</rdf:li> <rdf:li>255, 255</rd
f:li> </rdf:Seq> </crs:ToneCurvePV2012> <crs:ToneCurvePV2012Blue> <rdf:Seq> <rdf:li>0, 0</rdf:li> <rdf:li>255, 255</rdf:li> </rdf:Seq> </crs:ToneCurv
ePV2012Blue> <crs:ToneCurvePV2012Green> <rdf:Seq> <rdf:li>0, 0</rdf:li> <rdf:li>255, 255</rdf:li> </rdf:Seq> </crs:ToneCurvePV2012Green> <crs:ToneCur
vePV2012Red> <rdf:Seq> <rdf:li>0, 0</rdf:li> <rdf:li>255, 255</rdf:li> </rdf:Seq> </crs:ToneCurvePV2012Red> <lr:hierarchicalSubject> <rdf:Bag> <rdf:li
i>Schools|Merrick School of Business</rdf:li> </rdf:Bag> </lr:hierarchicalSubject> </rdf:Description> </rdf:RDF> </x:xmpmeta>

<?xpacket end="w"?>

Copyright 1999 Adobe Systems Incorporated
Adobe RGB (1998)
Adobe
Gimp
---- Full readable string count ----
21307
```

```
cd ~/CIP-DF-Lab3_yusuf_dada
strings evidence/J_ub_law.jpg | head -n 40
echo "---- Searching for common metadata indicators ----"
strings evidence/J_ub_law.jpg | grep -iE "canon|nikon|sony|samsung|apple|adobe|phot
oshop|gimp|software|camera|date|gps|exif" | tee hex_analysis/metadata_strings_found.
txt
```

Part C - Direct Data-Unit Examination with The Sleuth Kit

```
(kali@kali)~[~/CIP-DF-Lab3_yusuf_dada]
$ cd ~/CIP-DF-Lab3_yusuf_dada
img_stat evidence/Ch01InChap01.dd
mmls evidence/Ch01InChap01.dd
IMAGE FILE INFORMATION

Image Type: raw

Size in bytes: 1474560
Sector size: 512
```

```
cd ~/CIP-DF-Lab3_yusuf_dada
img_stat evidence/Ch01InChap01.dd
mmls evidence/Ch01InChap01.dd
```

```
(kali@kali)~[~/CIP-DF-Lab3_yusuf_dada]
$ fsstat evidence/Ch01InChap01.dd
fls -r -d evidence/Ch01InChap01.dd
FILE SYSTEM INFORMATION

File System Type: FAT12

OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: *3♦♦m|

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879

METADATA INFORMATION

Range: 2 - 45782
Root Directory: 2
```

```
CONTENT INFORMATION

Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 2848

FAT CONTENTS (in sectors)

33-236 (204) → EOF
285-311 (27) → EOF
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
```

```
fsstat evidence/Ch01InChap01.dd
fls -r -d evidence/Ch01InChap01.dd
```

Pick one deleted file from your fls output and examine/recover it directly

```
icat evidence/Ch01InChap01.dd 8 > carved/tsk_recovered_file
```

```
blkcat evidence/Ch01InChap01.dd 33 > carved/tsk_block_sample.dat
```

8 is on of the inode while 33 is block number I used

```
(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada]
$ icat evidence/Ch01InChap01.dd 8 > carved/tsk_recovered_file
blkcat evidence/Ch01InChap01.dd 33 > carved/tsk_block_sample.dat

(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada]
$ echo "--- Full readable string count ---"
strings evidence/J_ub_law.jpg | wc -l
--- Full readable string count ---
21307
```

```
echo "--- Full readable string count ---"
```

```
strings evidence/J_ub_law.jpg | wc -l
```

Part D - Embedded File Discovery with Binwalk

```
(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada]
$ binwalk evidence/J_ub_law.jpg
binwalk evidence/Ch01InChap01.dd

DECIMAL      HEXADECIMAL  DESCRIPTION
-----
0            0x0          JPEG image data, EXIF standard
12           0xC          TIFF image data, little-endian offset of first image directory: 8
26844        0x68DC       Copyright string: "Copyright 1999 Adobe Systems Incorporated"

DECIMAL      HEXADECIMAL  DESCRIPTION
-----
```

```
cd ~/CIP-DF-Lab3_yusuf_dada
```

```
binwalk evidence/J_ub_law.jpg
```

```
binwalk evidence/Ch01InChap01.dd
```

I used Binwalk to scan the files for known signatures **anywhere** inside it (not just at the very start)

```
(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada]
$ mkdir -p working
dd if=evidence/J_ub_law.jpg of=working/manual_carved_exif bs=1 skip=12
file working/manual_carved_exif
1692138+0 records in
1692138+0 records out
1692138 bytes (1.7 MB, 1.6 MiB) copied, 6.75477 s, 251 kB/s
working/manual_carved_exif: TIFF image data, little-endian, direntries=16, width=4928, height=3280, bps=206, PhotometricInterpretation=RGB, manufac-
rer=NIKON CORPORATION, model=NIKON D4, orientation=upper-left
```

```
Mkdir -p working
```

```
dd if=evidence/J_ub_law.jpg of=working/manual_carved_exif bs=1 skip=12
```

```
file working/manual_carved_exif
```


Part E - Prepare the USB Image

```
└─$ cd ~/CIP-DF-Lab3_yusuf_dada/evidence
7z x 120M.7z -o../working/
cd ../working
ls -lh

7-Zip 26.02 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-06-25
64-bit locale=en_US.UTF-8 Threads:32 OPEN_MAX:4096, ASM

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Extracting archive: 120M.7z
--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

Everything is Ok

Folders: 1
Files: 2
Size:      124782229
Compressed: 36720470
total 3.3M
drwxrwxr-x 2 kali kali 4.0K Sep 22  2021 120M
-rw-rw-r-- 1 kali kali 1.7M Sep  3 16:58 1_ub_low_reconstructed.jpg
-rw-rw-r-- 1 kali kali 1.7M Sep  3 19:15 manual_carved_exif
```

```
cd ~/CIP-DF-Lab3_yusuf_dada/evidence
7z x 120M.7z -o../working/
cd ../working
ls -lh
```

```
(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada/working]
└─$ ls -lh 120M/
total 120M
-rw-rw-r-- 1 kali kali 119M Sep 22  2021 usb_fat_carving.001
-rw-rw-r-- 1 kali kali 1.7K Sep 22  2021 usb_fat_carving.001.txt
```

Identify the extracted image file which usually end up with (a .dd, .img, or .raw file).

usb_fat_carving.001 is the extracted file from the outcome as seen

```
(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada/working/120M]
$ md5sum usb_fat_carving.001 | tee ../report/usb_image_md5.txt
sha256sum usb_fat_carving.001 | tee ../report/usb_image_sha256.txt
img_stat usb_fat_carving.001
mmls usb_fat_carving.001
tee: ../report/usb_image_md5.txt: No such file or directory
ba4a1d0ba49f4a6667b00a3b3e85e604 usb_fat_carving.001
tee: ../report/usb_image_sha256.txt: No such file or directory
9bfe4b5634ade30764f0e581a4686930f7fab0472378595b789b0cdb248c91d9 usb_fat_carving.001
IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 124780544
Sector size: 512
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

    Slot      Start      End      Length    Description
000:  Meta      0000000000  0000000000  0000000001  Primary Table (#0)
001:  _____ 0000000000  0000000127  0000000128  Unallocated
002:  000:000  0000000128  0000243711  0000243584  Win95 FAT16 (0x0e)
```

```
md5sum usb_fat_carving.001 | tee ../report/usb_image_md5.txt
sha256sum usb_fat_carving.001 | tee ../report/usb_image_sha256.txt
img_stat usb_fat_carving.001
mmls usb_fat_carving.001
```

It shows extraction, hashes, img_stat, and mmls output with my confirmed offset clearly visible

```
(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada/working/120M]
$ fsstat -o 128 usb_fat_carving.001
FILE SYSTEM INFORMATION
-----
File System Type: FAT16

OEM Name: MSDOS5.0
Volume ID: 0x94105672
Volume Label (Boot Sector): NO NAME
Volume Label (Root Directory): USB
File System Type Label: FAT16

Sectors before file system: 128

File System Layout (in sectors)
Total Range: 0 - 243583
* Reserved: 0 - 3
** Boot Sector: 0
* FAT 0: 4 - 241
* FAT 1: 242 - 479
* Data Area: 480 - 243583
** Root Directory: 480 - 511
** Cluster Area: 512 - 243583

METADATA INFORMATION
-----
Range: 2 - 3889670
Root Directory: 2
```

```

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 2048
Total Cluster Range: 2 - 60769

FAT CONTENTS (in sectors)
-----
512-515 (4) → EOF
516-519 (4) → EOF
520-523 (4) → EOF
70976-71211 (236) → EOF
71212-71215 (4) → EOF

```

fsstat -o 128 usb_fat_carving.001

Part F - Carve Deleted Files with Scalpel

F1. Configure Scalpel

```

GNU nano 9.2 /etc/scalpel/scalpel.conf
# are used. No footer is defined and the max carve size is 1000 bytes.
#
# NONE y 1000 FOREMOST
#
# GRAPHICS FILES
#
#
# AOL ART files
# art y 150000 \x4a\x47\x04\x0e \xcf\xcb\xcb
# art y 150000 \x4a\x47\x03\x0e \xd0\xcb\x00\x00
#
# GIF and JPG files (very common)
# gif y 5000000 \x47\x49\x46\x38\x37\x61 \x00\x3b
# gif y 5000000 \x47\x49\x46\x38\x39\x61 \x00\x3b
# jpg y 5242880 \xff\xd8\xff???Exif \xff\xd9 REVERSE
# jpg y 5242880 \xff\xd8\xff???JFIF \xff\xd9 REVERSE
#
# PNG
# png y 20000000 \x50\x4e\x47? \xff\xfc\xfd\xfe
#
# BMP (used by MSWindows, use only if you have reason to think there are
# BMP files worth digging for. This often kicks back a lot of false
# positives
#
#
# Help Write Out Where Is Cut Execute Location M-U Undo M-A Set Mark
# Exit Read File Replace Paste Justify Go To Line M-E Redo M-C Copy

```

sudo nano /etc/scalpel/scalpel.conf

Find the lines for JPEG (they'll look like this, commented out with a leading #):

```

JPEG
jpg y 200000000 \xff\xd8\xff\xe0\x00\x10 \xff\xd9
jpg y 200000000 \xff\xd8\xff\xe1 \xff\xd9

```

F2. Run Scalpel against the USB image

```
(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada]
$ cd ~/CIP-DF-Lab3_yusuf_dada
sudo scalpel -o scalpel_output/ working/120M/usb_fat_carving.001
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/kali/CIP-DF-Lab3_yusuf_dada/working/120M/usb_fat_carving.001"

Image file pass 1/2.
working/120M/usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Allocating work queues...
Work queues allocation complete. Building carve lists...
Carve lists built. Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x45\x78\x69\x66" and footer "\xff\xd9" → 8 files
jpg with header "\xff\xd8\xff\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" → 9 files
Carving files from image.
Image file pass 2/2.
working/120M/usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Processing of image file complete. Cleaning up...
Done.
Scalpel is done, files carved = 17, elapsed = 6 seconds.
```

```
cd ~/CIP-DF-Lab3_yusuf_dada
sudo scalpel -o scalpel_output/ working/120M/usb_fat_carving.001
```

The Scalpel run completing, showing how many files of each type were carved.

Scalpel is known to scan the entire image byte-by-byte for JPEG headers/footers and carve out anything matching, regardless of file-system metadata.

F3. Review the results

```
(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada]
$ ls -la scalpel_output/
cat scalpel_output/audit.txt
ls: cannot access 'scalpel_output/jpg-1-0': Permission denied
ls: cannot access 'scalpel_output/jpg-0-0': Permission denied
ls: cannot access 'scalpel_output/audit.txt': Permission denied
ls: cannot access 'scalpel_output/.': Permission denied
ls: cannot access 'scalpel_output/..': Permission denied
total 0
d???????? ? ? ? ? ? ? ? ?
d???????? ? ? ? ? ? ? ? ?
-???????? ? ? ? ? ? ? ? ?
d???????? ? ? ? ? ? ? ? ?
d???????? ? ? ? ? ? ? ? ?
cat: scalpel_output/audit.txt: Permission denied
```

```
(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada]
$ sudo ls -la scalpel_output/
cat scalpel_output/audit.txt
total 20
drwxr-xr-- 4 root root 4096 Sep  4 07:23 .
drwxrwxr-x 8 kali kali 4096 Sep  4 07:22 ..
-rw-r--r-- 1 root root 1618 Sep  4 07:23 audit.txt
drwxr-xr-x 2 root root 4096 Sep  4 07:23 jpg-0-0
drwxr-xr-x 2 root root 4096 Sep  4 07:23 jpg-1-0
cat: scalpel_output/audit.txt: Permission denied
```

```
Sudo ls -la scalpel_output/
cat scalpel_output/audit.txt
```

The audit.txt file lists every carved file, its size, and the byte offset it was found at, this is Scalpel's own evidence log, and it's specifically required.


```
(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada]
$ sudo find scalpel_output/ -name "*.jpg" -o -name "*.jpeg"
scalpel_output/jpg-1-0/00000009.jpg
scalpel_output/jpg-1-0/00000010.jpg
scalpel_output/jpg-1-0/00000014.jpg
scalpel_output/jpg-1-0/00000011.jpg
scalpel_output/jpg-1-0/00000012.jpg
scalpel_output/jpg-1-0/00000008.jpg
scalpel_output/jpg-1-0/00000016.jpg
scalpel_output/jpg-1-0/00000013.jpg
scalpel_output/jpg-1-0/00000015.jpg
scalpel_output/jpg-0-0/00000000.jpg
scalpel_output/jpg-0-0/00000004.jpg
scalpel_output/jpg-0-0/00000006.jpg
scalpel_output/jpg-0-0/00000003.jpg
scalpel_output/jpg-0-0/00000005.jpg
scalpel_output/jpg-0-0/00000002.jpg
scalpel_output/jpg-0-0/00000007.jpg
scalpel_output/jpg-0-0/00000001.jpg
```

`Sudo find scalpel_output/ -name "*.jpg" -o -name "*.jpeg"`

audit.txt contents and the list of recovered JPEG files.

F4. Verify at least two recovered JPEGs actually open

```
(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada]
$ sudo find scalpel_output/ -iname "*.jpg" | head -n 5
# Open at least two of them:
eog scalpel_output/jpg-1-0/00000008.jpg 2>/dev/null || xdg-open scalpel_output/jpg-1-0/00000009.jpg
scalpel_output/jpg-1-0/00000009.jpg
scalpel_output/jpg-1-0/00000010.jpg
scalpel_output/jpg-1-0/00000014.jpg
scalpel_output/jpg-1-0/00000011.jpg
scalpel_output/jpg-1-0/00000012.jpg
^C
```

`find scalpel_output/ -iname "*.jpg" | head -n 5`

Open at least two of them:

`eog scalpel_output/jpg-1-0/00000008.jpg 2>/dev/null || xdg-open scalpel_output/jpg-1-0/00000009.jpg`

F5. Hash the recovered files

```
(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada]
$ cd scalpel_output
sudo find . -iname "*.jpg" -exec sha256sum {} \; | tee ../report/carved_jpegs_sha256.txt
cd: permission denied: scalpel_output
tee: ../report/carved_jpegs_sha256.txt: No such file or directory
[sudo] password for kali:
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 ./evidence/J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 ./working/J_ub_law_reconstructed.jpg
5cf67210983c08d8800e46cbd771b22b4a6bb0cccf96130b2106d6845094b06 ./scalpel_output/jpg-1-0/00000009.jpg
7c127f3b90c3a0052cc019392ef4bb5ec17d7ec8a39fdd17d01ccd6a58be49ab ./scalpel_output/jpg-1-0/00000010.jpg
eb939a9b48da622896cf8577e9ccbd1c38a4b1686f09b3512282942d0c826411 ./scalpel_output/jpg-1-0/00000014.jpg
dbf6aa689db8256e8d78cfb5d2016a66ee46dfc40690f9f1acb8d467e673d4ee ./scalpel_output/jpg-1-0/00000011.jpg
371807a9173648494a5ec4c3d63739ae6adeaab3646c15e29421eb64d46fbee4 ./scalpel_output/jpg-1-0/00000012.jpg
3eeeb18dbaa50a6ae3cbb7630bd7fe426b317eb79fc951f3a6ceb601f8639d1 ./scalpel_output/jpg-1-0/00000008.jpg
5b7a35a6b0bd67deaa1dbd92ee6b31950ace908c44f5124639d180a5e40e8567 ./scalpel_output/jpg-1-0/00000016.jpg
39a81569027ab59f0febe2bc518c6e519b763fb8ccd177b86841aaf84239b5a1 ./scalpel_output/jpg-1-0/00000013.jpg
3d31fb0df55a4558446f8577cb7943a7dbd9fe44f782fdc8d64b13c4120d89d4 ./scalpel_output/jpg-1-0/00000015.jpg
df36e6bcb12c782ab67a6662b5d8d306383cfe67ca050485a0f97d2db1aff293 ./scalpel_output/jpg-0-0/00000000.jpg
```

cd scalpel_output

sudo find . -iname "*.jpg" -exec sha256sum {} \; | tee ../report/carved_jpegs_sha256.txt

Final Step - Submission analysis in dorensics

```
(kali@kali)-[~/CIP-DF-Lab3_yusuf_dada]
$ cd ~/CIP-DF-Lab3_yusuf_dada
zip -r CIP-DF-Lab3_yusuf_dada_Submission.zip \
  hex_analysis working carved scalpel_output report
ls -lh CIP-DF-Lab3_yusuf_dada_Submission.zip
      zip warning: name not matched: scalpel_output/jpg-1-0
      zip warning: name not matched: scalpel_output/jpg-0-0
      zip warning: name not matched: scalpel_output/audit.txt
  adding: hex_analysis/ (stored 0%)
  adding: hex_analysis/metadata_strings_found.txt (deflated 77%)
  adding: hex_analysis/J_ub_low_hexdump.txt (deflated 48%)
  adding: working/ (stored 0%)
  adding: working/manual_carved_exif (deflated 1%)
  adding: working/120M/ (stored 0%)
  adding: working/120M/usb_fat_carving.001 (deflated 70%)
  adding: working/120M/usb_fat_carving.001.txt (deflated 56%)
  adding: working/J_ub_low_reconstructed.jpg (deflated 1%)
  adding: carved/ (stored 0%)
  adding: carved/tsk_block_sample.dat (deflated 69%)
  adding: carved/tsk_recovered_file (deflated 88%)
  adding: scalpel_output/ (stored 0%)
  adding: report/ (stored 0%)
  adding: report/usb_image_sha256.txt (stored 0%)
  adding: report/evidence_md5.txt (deflated 17%)
  adding: report/usb_image_md5.txt (stored 0%)
  adding: report/evidence_sha256.txt (deflated 27%)
-rw-rw-r-- 1 kali kali 43M Sep  4 13:08 CIP-DF-Lab3_yusuf_dada_Submission
.zip
```

```
cd ~/CIP-DF-Lab3_yusuf_dada
zip -r CIP-DF-Lab3_yusuf_dada_Submission.zip \
  hex_analysis working carved scalpel_output report
ls -lh CIP-DF-Lab3_yusuf_dada_Submission.zip
```